LESSON 01 / 08

What a^x means (and when it exists)

An **exponent** a^x answers "multiply a by itself x times" only when x is a positive integer. Every other exponent is an *extension* built to keep the laws below true:

- $a^0 = 1$ (empty product) for every $a \neq 0$.
- $a^{-p} = \dfrac{1}{a^p}$ for $p > 0$.
- $a^{p/q} = (\sqrt[q]{a})^p = (a^{1/q})^p$ for $p \in \mathbb{Z}$, $q \in \mathbb{N}$, $q \geq 1$, $\gcd(p, q) = 1$.

Base convention (the single most important assumption of the chapter): for *real* exponents x that are not integers, a^x is defined for $a > 0$ only. With $a > 0$, a^x exists for every real x and is positive.

> **ASSUMPTIONS & CONVENTIONS**
> Throughout this chapter a base written a, b, c under a real exponent is assumed **positive**. If the base can be negative, only *integer* exponents are safe, and rational exponents need the parity rule below. Logarithm bases further satisfy $a \neq 1$. When a problem writes a^x with no comment, $a > 0$ is part of the data.

Rational exponents and parity

For $a > 0$: $a^{p/q}$ is the **unique positive** q-th root raised to p. For $a < 0$ the real q-th root exists **iff** q **is odd**. So $(-8)^{1/3} = -2$ is legal, while $(-8)^{1/2}$ is not real.

$(-8)^{2/3} = ((-8)^{1/3})^2 = (-2)^2 = 4$ — reduce p/q to lowest terms **first**, then take the odd root.

The laws of indices (and their hidden hypotheses)

For $a, b > 0$ and $r, s \in \mathbb{R}$:

$$a^r a^s = a^{r+s}, \qquad \frac{a^r}{a^s} = a^{r-s}, \qquad (a^r)^s = a^{rs},$$

$$(ab)^r = a^r b^r, \qquad \left(\frac{a}{b}\right)^r = \frac{a^r}{b^r}, \qquad a^{-r} = \frac{1}{a^r}.$$

> **SHORT DERIVATION**
> **Why $(ab)^r = a^r b^r$ needs $a, b > 0$**
> For integer r the identity holds for all reals a, b (just expand). For $r = 1/2$ it fails on negatives: $((-1)(-1))^{1/2} = 1^{1/2} = 1$, but $(-1)^{1/2} \cdot (-1)^{1/2}$ is not real. The clean repair: restrict to $a, b > 0$, where every factor a^r is a well-defined positive number. That is exactly the convention above.

COMMON TRAP

Common trap — $(a^r)^s = a^{rs}$ is false on negatives

$((-2)^2)^{1/2} = 4^{1/2} = 2$, but $(-2)^{2\cdot 1/2} = (-2)^1 = -2$. The rule $(a^r)^s = a^{rs}$ is valid for $a > 0$ (or when r, s are both integers). Mixing "even power" and "half power" on a negative base is the classic JEE trap. Two safe forms: $(a^2)^{1/2} = |a|$ for all real a; $a^{1/2}$ only for $a \geq 0$.

The absolute-value repairs

$$\sqrt{x^2} = |x| \text{ for all real } x; \qquad (x^{1/2})^2 = x \text{ for } x \geq 0$$

$$\sqrt[n]{x^n} = |x| \text{ if } n \text{ even}; \quad x \text{ if } n \text{ odd}$$

KEY RESULT

Three repairs worth memorising: (1) $\sqrt{x^2} = |x|$, never x; (2) reduce p/q to lowest terms before evaluating a negative base; (3) $(a^r)^s = a^{rs}$ and $(ab)^r = a^r b^r$ are stated here for $a, b > 0$ — that is the standing convention.

Solving exponent equations on the level of indices

If $a > 0$, $a \neq 1$, then $a^{f(x)} = a^{g(x)} \Leftrightarrow f(x) = g(x)$ — the exponential function is **injective** (one-to-one). If $a = 1$, both sides are 1 and the equation is empty of information; if $a = 0$ or the base can vary, handle cases.

WORKED EXAMPLE

Worked. Solve $x^{2/3} = 8$ over $\mathbb{R}$

Raise both sides to the power $3/2$ with care. Write $x^{2/3} = (x^{1/3})^2 = 8$, so $x^{1/3} = \pm 2\sqrt{2}$ and $x = \pm(2\sqrt{2})^3 = \pm 16\sqrt{2}$. Both are valid (the exponent $2/3$ has odd denominator, so $(-16\sqrt{2})^{2/3} = ((-16\sqrt{2})^{1/3})^2 = (-2\sqrt{2})^2 = 8$ ✓). Answer: $x = \pm 16\sqrt{2}$.

COMMON TRAP

Common trap — squaring and simplifying add or drop roots

$x^2 = 4 \Rightarrow x = \pm 2$, not $x = 2$. But $x^{1/2} = 2 \Rightarrow x = 4$ **only** (the left side forces $x \geq 0$ and is already non-negative; squaring here does not add roots). Before any manipulation, write the **domain** of the equation; after, check every candidate in the original equation.

LESSON 02 / 08

a^x as a function $\mathbb{R} \to (0, \infty)$

Fix $a > 0$. Then $f(x) = a^x$ has **domain** $\mathbb{R}$, **range** $(0, \infty)$, and satisfies the **exponential law**

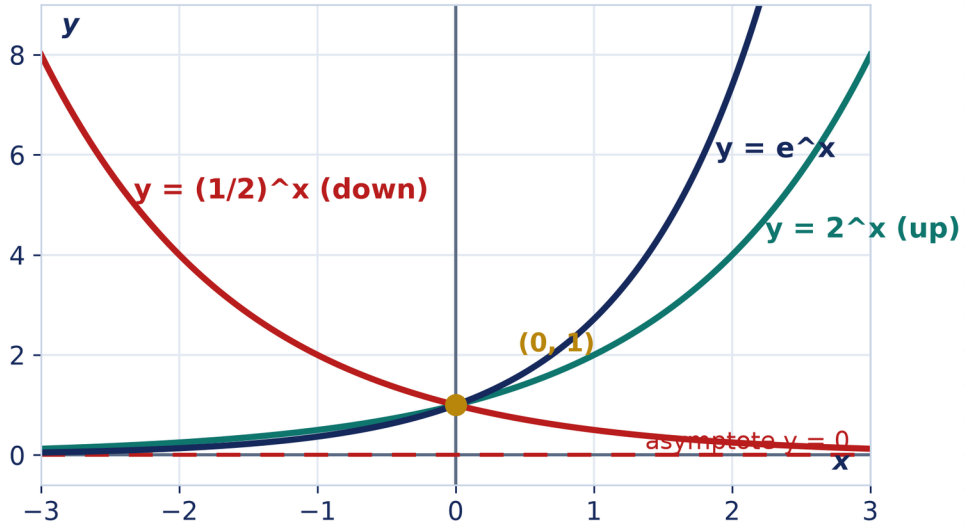
$$f(x+y) = f(x)f(y), \qquad f(0) = 1, \qquad f(-x) = \frac{1}{f(x)}.$$

(Any continuous function with $f(x+y) = f(x)f(y)$ and $f(1) = a > 0$ must be a^x — the law plus continuity pins the whole function down. Discontinuous counter-examples exist but need set-theoretic constructions beyond this syllabus; exam functions are continuous.)

- $a > 1$: strictly **increasing**, $a^x \to \infty$ as $x \to \infty$, $a^x \to 0^+$ as $x \to -\infty$.
- $0 < a < 1$: strictly **decreasing**, $a^x \to 0^+$ as $x \to \infty$, $a^x \to \infty$ as $x \to -\infty$.
- $a = 1$: the constant 1 — which is why log bases exclude 1.

Every curve $y = a^x$ passes through $(0, 1)$ and lies **above the x-axis**; the x-axis is a horizontal asymptote.

The exponential family (all pass through (0, 1), asymptote y = 0)



The number e and the natural exponential

$e = 2.718281828459045\ldots$ is the unique base for which the curve $y = e^x$ has slope 1 at $x = 0$ (the fact that will make e^x the natural exponential in calculus). Facts used constantly:

$$e \approx 2.71828, \qquad \ln e = 1, \qquad e^{x+y} = e^x e^y$$

$$(e^x)^r = e^{rx}$$

Notation: $\lg x = \log_{10} x$ (common log), $\ln x = \log_e x$ (natural log), and $a^x = e^{x \ln a}$ for every $a > 0$ — this identity converts any exponential base into e and is the workhorse of comparisons.

KEY RESULT

Monotonicity is the whole game. For $a > 1$: $a^{f(x)} > a^{g(x)} \Leftrightarrow f(x) > g(x)$. For $0 < a < 1$ the inequality **flips**: $a^{f(x)} > a^{g(x)} \Leftrightarrow f(x) < g(x)$. Same rule for log inequalities in the next sheets.

SHORT DERIVATION

Why a^x is increasing when $a > 1$ (algebraic proof)
Let $x < y$, so $y - x > 0$. Then $a^y = a^x \cdot a^{y-x}$. For $a > 1$ and $h > 0$ we have $a^h > 1$ (the q-th root of a number > 1 is > 1, and rationals are dense / continuity), so $a^y > a^x$. For $0 < a < 1$, $a^h < 1$ and the inequality reverses. This is the entire content of "exp preserves inequalities" — one multiplication by a factor > 1 or < 1.

Standard applications

1. Exponential equations by substitution. $4^x - 3 \cdot 2^x + 2 = 0$: put $t = 2^x > 0$, get $t^2 - 3t + 2 = 0$, so $t \in \{1, 2\}$, i.e. $x \in \{0, 1\}$. Always keep the constraint $t > 0$ in the quadratic stage.

2. Growth comparisons. To compare 2^{300} and 3^{200}, take lg: $300 \lg 2 \approx 90.31$ and $200 \lg 3 \approx 95.42$, so $3^{200} > 2^{300}$. Taking a log of a positive comparison never changes its direction.

3. Mixed systems. Solve $e^{2x} - 3e^x + 2 = 0$ by $t = e^x$: $t = 1, 2$, so $x = 0$ or $x = \ln 2$.

WORKED EXAMPLE

Worked. Solve $2^{2x} - 5 \cdot 2^x + 4 = 0$
$t = 2^x > 0$: $t^2 - 5t + 4 = 0 \Rightarrow t = 1$ or $t = 4$ (both > 0 ✓). So $2^x = 1 \Rightarrow x = 0$; $2^x = 4 \Rightarrow x = 2$. Check $x = 0$: $1 - 5 + 4 = 0$ ✓; $x = 2$: $16 - 20 + 4 = 0$ ✓. Answer: $x \in \{0, 2\}$.

COMMON TRAP

Common traps — exponential edition
- $(a^x)^2 = a^{2x}$ ✓ for $a > 0$; but $a^x = a^y \Rightarrow x = y$ only when $a \neq 1$.
- Substituting $t = a^x$ **throws away** the range condition $t > 0$: a quadratic giving $t = -2$ contributes nothing.
- $2^x = -4$ has **no real solution** (range of 2^x is $(0, \infty)$) — say so instead of writing $x = 2i$.
- $a^x > 0$ always: $a^x \geq 0$ or $a^x > c$ for $c < 0$ is automatically true for all x.

LESSON 03 / 08

The definition (memorise the equivalence)

For $a > 0$, $a \neq 1$ and $y > 0$:

$$\log_a y = x \quad \Longleftrightarrow \quad a^x = y.$$

Read it both ways: $\log_a y$ is the **exponent** to which a must be raised to get y. Immediate consequences (all just the definition in disguise):

$$\log_a 1 = 0, \qquad \log_a a = 1, \qquad \log_a \frac{1}{a} = -1$$

$$\log_a a^p = p \quad (\forall p \in \mathbb{R})$$

$$a^{\log_a y} = y \quad (y > 0)$$

Domain discipline: $\log_a y$ requires $y > 0$ **and** $a > 0$, $a \neq 1$. $\log_a 0$ and $\log_a(\text{negative})$ are undefined in $\mathbb{R}$; $\log_1 y$ and $\log_{-2} y$ are meaningless (no unique inverse of 1^x or of $(-2)^x$).

The three laws (each comes from a law of indices)

For $M, N > 0$, $a > 0$, $a \neq 1$, $p \in \mathbb{R}$:

$$\log_a (MN) = \log_a M + \log_a N$$

$$\log_a \left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a (M^p) = p \log_a M.$$

SHORT DERIVATION

Derivation — set $M = a^m$, $N = a^n$

Then $MN = a^{m+n}$, so by definition $\log_a (MN) = m + n = \log_a M + \log_a N$. The quotient law is the same with $N \mapsto 1/N$. For the power law, $M^p = (a^m)^p = a^{pm}$ (here $M > 0$ makes $(a^m)^p = a^{mp}$ legitimate), so $\log_a (M^p) = pm$. Every log law is an index law in disguise — that is why the domain restrictions above are non-negotiable.

COMMON TRAP

Common traps — there is no law for sums or products of logs

- $\log_a(M+N)$ **does not expand**. There is no formula. Expand $\log(MN)$ or $\log(M/N)$ only.
- $\log_a M \cdot \log_a N$ is not the log of anything.
- $(\log_a M)^p \neq \log_a(M^p)$ in general: $p\log_a M = \log_a(M^p)$, but $(\log_a M)^p$ is just a power of a number.
- $\log_a(x^2) = 2\log_a x$ is **wrong for** $x<0$ — correct form: $\log_a(x^2) = 2\log_a|x|$ for $x \neq 0$.
- $\log_a a^p = p$ holds for all real p; $a^{\log_a x} = x$ holds only for $x>0$.

Change of base, chain rule, reciprocity

For $a, b, c > 0$, $a, b \neq 1$:

$$\log_a b = \frac{\log_c b}{\log_c a} = \frac{\ln b}{\ln a} = \frac{\lg b}{\lg a}$$

$$\log_a b \cdot \log_b a = 1, \qquad \log_a b \cdot \log_b c = \log_a c.$$

SHORT DERIVATION

Derivation — put $x = \log_a b$, **so** $a^x = b$

Take $\log_c$ of both sides: $x\log_c a = \log_c b$, hence $x = \frac{\log_c b}{\log_c a}$. Setting $c = b$ gives $\log_a b = 1/\log_b a$ (the reciprocity $\log_a b \cdot \log_b a = 1$); chaining two such fractions gives $\log_a b \cdot \log_b c = \frac{\ln b}{\ln a} \cdot \frac{\ln c}{\ln b} = \frac{\ln c}{\ln a} = \log_a c$.

The sign rule and log inequalities

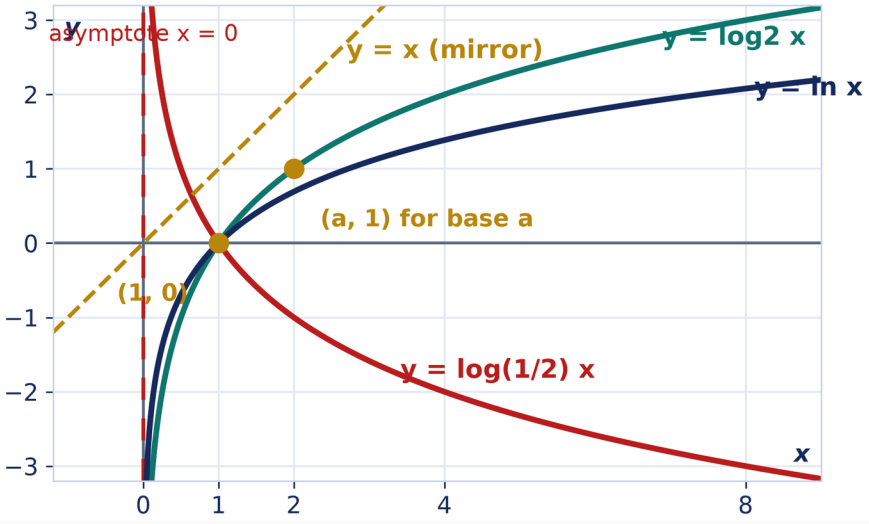
For $a > 0$, $a \neq 1$, $x > 0$:

$$\log_a x > 0 \iff (x-1)(a-1) > 0$$

(equivalently: x and a lie on the same side of 1)

So $\log_2 5 > 0$, $\log_{1/2} 5 < 0$, $\log_2 0.3 < 0$. Monotonicity: $\log_a x$ is **increasing** for $a > 1$, **decreasing** for $0 < a < 1$ — so $\log_a M > \log_a N \Rightarrow M > N$ for $a > 1$ but $\Rightarrow M < N$ for $0 < a < 1$.

Log curves (all pass (1, 0)), vertical asymptote x = 0, reflection in y = x



KEY RESULT

*The graph of $y = \log_a x$ is the reflection of $y = a^x$ in the line $y = x$ (inverse functions). Both pass $(1, 0)$ / $(0, 1)$ respectively; log has **vertical** asymptote $x = 0$, exp has **horizontal** asymptote $y = 0$; range of log = domain of exp = $\mathbb{R}$, domain of log = range of exp = $(0, \infty)$.*

WORKED EXAMPLE

Worked. Simplify $\log_2 6 - \log_2 3 + \log_2 16$

$= \log_2\left(\frac{6}{3} \cdot 16\right) = \log_2 32 = 5$. *Check:* $2^5 = 32 = 2 \cdot 16$ ✓.

LESSON 04 / 08

Simplification patterns that recur in exams

Nested logs peel from the inside: $\log_2(\log_3 81) = \log_2(4) = 2$. **Mixed-base sums** use change of base: $\log_4 8 = \dfrac{\log_2 8}{\log_2 4} = \dfrac{3}{2}$.

Power towers of a single base: $\log_a(a^{a^x}) = a^x$ (one application of $\log_a(a^p) = p$).

Sums of logs at fractions:
$\log_2 3 + \log_2 6 + \log_2 24 = \log_2(3 \cdot 6 \cdot 24) = \log_2 432 = \log_2(2^4 \cdot 27) = 4 + 3\log_2 3$.

SHORT DERIVATION

The digit formula — how many digits does N have?
A positive integer N has d digits exactly when $10^{d-1} \le N < 10^d$. Taking lg: $d - 1 \le \lg N < d$, so $d = \lfloor \lg N \rfloor + 1$. Example: 2^{100}. $\lg(2^{100}) = 100\lg 2 \approx 30.103$, so 2^{100} has $\lfloor 30.103 \rfloor + 1 = 31$ digits ✓ (indeed $2^{100} = 1267650600228229401496703205376$). The formula uses $\lfloor \cdot \rfloor$, **not rounding**, and works for any positive real N ("order of magnitude" version).

Comparing magnitudes — the three standard moves

1. **Take a log.** 2^{300} vs 3^{200}: compare $300\lg 2 \approx 90.31$ vs $200\lg 3 \approx 95.42 \Rightarrow 3^{200} > 2^{300}$. Logs are increasing, so log of two positives preserves their order (use ln or lg, any base > 1).

1. **Raise to a common power.** a^b vs b^a with $a < b$: compare after raising both to $1/(ab)$: $a^b < b^a \Leftrightarrow a^{1/a} < b^{1/b} \Leftrightarrow \dfrac{\ln a}{a} < \dfrac{\ln b}{b}$. The function $f(t) = \dfrac{\ln t}{t}$ increases up to $t = e$ and decreases after (calculus confirms; for the toolkit, know the shape: max at e).

1. **Squeeze between integers.** For integer estimates, bracket with known powers: $2^{10} = 1024 \approx 10^{3.01}$ is the usual bridge ($1000 \approx 2^{10}$).

WORKED EXAMPLE

Worked. Which is larger: e^π or π^e?
Take ln of both (both positive, ln increasing): $\ln(e^\pi) = \pi \approx 3.14159$, while $\ln(\pi^e) = e\ln\pi \approx 2.71828 \times 1.14473 \approx 3.11229$. Since $3.14159 > 3.11229$, we get $e^\pi > \pi^e$. (Consistent with $f(t) = \frac{\ln t}{t}$ decreasing for $t > e$: here $\pi > e$, so $\frac{\ln e}{e} > \frac{\ln \pi}{\pi} \Rightarrow \pi \ln e > e \ln \pi \Rightarrow e^\pi > \pi^e$ ✓.)

Sign and size of $\log_a x$ — the grid to hold in your head

	$x > 1$	$x = 1$	$0 < x < 1$
$a > 1$	$\log_a x > 0$	0	$\log_a x < 0$
$0 < a < 1$	$\log_a x < 0$	0	$\log_a x > 0$

Absolute value: $|\log_a x| = -\log_a x$ *exactly when* $\log_a x \le 0$, *i.e. when* x *and* a *sit on opposite sides of* 1. *So* $\log_a |x|$ *(defined for* $x \neq 0$*) and* $|\log_a x|$ *(defined for* $x > 0$*) are* ***different functions*** — *check which one a problem asks for.*

KEY RESULT

Comparison checklist: *(1) take* $\ln$ *of both sides if both are positive; (2) convert to a common base or to* e *via* $a^x = e^{x \ln a}$; *(3) for* a^b *vs* b^a *use* $f(t) = \frac{\ln t}{t}$ *(max at* $t = e$; $2^4 = 4^2$ *is the only integer pair with* $a \neq b$ *and* $a^b = b^a$, *namely* $(2, 4)$ *and* $(4, 2)$*); (4) remember* $\lg 2 \approx 0.301$, $\lg 3 \approx 0.477$, $\ln 2 \approx 0.693$ — *three numbers that unlock most numerical estimates.*

COMMON TRAP

Common traps — comparison edition

- *Taking* $\log$ *of an inequality requires* ***both sides positive***; $\log$ *of a negative is undefined, not "negative log".*
- $\lg(a + b) \neq \lg a + \lg b$; *write* $\lg(a + b)$ *as it is or factor first.*
- $\log_a x$ *increasing is for* $a > 1$ *only; with* $0 < a < 1$, $\log_a x > \log_a y \Rightarrow x < y$.
- $2^{300} \approx 10^{90.3}$ *means* 2^{300} *is between* 10^{90} *and* 10^{91} — *use floors, not rounding, for digits.*

LESSON 05 / 08

The four structural types

Type 1 — common base. $a^{f(x)} = a^{g(x)}$ with $a > 0$, $a \neq 1 \Rightarrow f(x) = g(x)$. *If the bases differ but are related ($4^x = 8^{x-1}$), convert:* $2^{2x} = 2^{3x-3} \Rightarrow x = 3$.

Type 2 — quadratic in an exponential. $a^{2x} + Ba^x + C = 0$: *substitute* $t = a^x > 0$, *solve, then reject* $t \leq 0$ *and take* $x = \log_a t$.

Type 3 — logarithmic equation to exponential. *Isolate a single log, then rewrite:* $\log_2(x-3) = 4 \Rightarrow x - 3 = 2^4 \Rightarrow x = 19$. *If logs remain, use laws to* **collapse** *to one log first:* $\log_2(x-3) + \log_2(x+1) = 3 \Rightarrow \log_2((x-3)(x+1)) = 3$.

Type 4 — x in base and argument. $x^{\log_2 x} = 16$*-type: take* $\log_2$*:* $(\log_2 x)^2 = 4 \Rightarrow \log_2 x = \pm 2 \Rightarrow x = 4$ *or* $x = \frac{1}{4}$. *Both check out (a power* x^y *with* $x > 0$ *is always legal).*

Domain checks turn 'solutions' into verified roots — worked example 2: only x = 2 + 2√3 survives



The domain-first discipline

Every log equation has an **implicit domain**: *each* $\log(\cdot)$ *needs its argument* > 0; *every* a^x *is fine on* $\mathbb{R}$ *for* $a > 0$; $x^{p/q}$ *needs the parity care of sheet 1. Solve* **first** *on the domain, then verify — or solve freely and* **filter at the end**. *Either way the final answer must be checked in the original equation, because squaring, exponentiating, and multiplying by logs can create ghosts.*

SHORT DERIVATION

Worked 1. $2^{2x} - 5 \cdot 2^x + 4 = 0$ **(type 2)**

$t = 2^x$, $t > 0$: $t^2 - 5t + 4 = 0 \Rightarrow t = 1, 4$, *both admissible.* $2^x = 1 \Rightarrow x = 0$; $2^x = 4 \Rightarrow x = 2$. *Substitute back:* $x = 0$ *gives* $1 - 5 + 4 = 0$ ✓; $x = 2$ *gives* $16 - 20 + 4 = 0$ ✓. *Answer* $\{0, 2\}$.

SHORT DERIVATION

Worked 2. $\log_2(x-3) + \log_2(x+1) = 3$ **(type 3)**

Domain: $x > 3$ **and** $x > -1$, *so* $x > 3$. *Collapse:* $\log_2((x-3)(x+1)) = 3 \Rightarrow (x-3)(x+1) = 8 \Rightarrow x^2 - 2x - 11 = 0 \Rightarrow x = 1 \pm 2\sqrt{3}$. *Only* $1 + 2\sqrt{3} \approx 4.464$ *lies in* $x > 3$; $1 - 2\sqrt{3} < 0$ *is a ghost created by the algebra (it would make the original logs undefined). Answer:* $x = 1 + 2\sqrt{3}$.

WORKED EXAMPLE

Worked 3. $x^{\lg x} = 10^4$ **(type 4)**

Take $\lg$ of both sides (both positive):

$(\lg x)(\lg x) = 4 \Rightarrow (\lg x)^2 = 4 \Rightarrow \lg x = \pm 2 \Rightarrow x = 100$ or $x = 10^{-2} = 0.01$. Check:

$100^{\lg 100} = 100^2 = 10^4$ ✓; $(0.01)^{\lg 0.01} = (10^{-2})^{-2} = 10^4$ ✓. Both answers valid.

COMMON TRAP

Common traps — equations edition

- **Collapsing logs** $\log A + \log B = \log(AB)$ requires $A, B > 0$ — the domain shrinks, it does not grow.
- Squaring both sides (e.g. when clearing $\sqrt{\cdot}$ or $|\cdot|$) creates ghost roots — verify each one.
- $\log_a f = \log_a g \Rightarrow f = g$ **plus** $f, g > 0$; never cancel logs before writing the domain.
- Exponentiating $e^A = e^B$ is reversible for all real A, B; but $A^2 = B^2 \Rightarrow A = \pm B$ — do not "cancel squares".
- If a base depends on x ($x^{\log x}$, $\log_x 2$), add the conditions $x > 0$, $x \neq 1$ (base rule) on top of argument rules.

LESSON 06 / 08

Log and exponent inequalities

Solve like equations, then respect the monotonicity direction. Worked: $\lg(x-1)<1$. Domain $x>1$; $\lg$ is increasing ($10>1$), so $x-1<10^1=10$; combine: $1<x<11$.

With a base <1 the inequality flips: $\log_{1/2}(x-1)<1 \Rightarrow x-1>(1/2)^1=\frac{1}{2}$ (flip!) $\Rightarrow x>\frac{3}{2}$.

Exponential inequalities follow the same rule: $2^{x^2-1}\geq 2^{3x} \Rightarrow x^2-1\geq 3x$ (base $2>1$) $\Rightarrow x^2-3x-1\geq 0 \Rightarrow x\leq \frac{3-\sqrt{13}}{2}$ or $x\geq \frac{3+\sqrt{13}}{2}$.

KEY RESULT

Inequality protocol: (1) write the domain first (logs: argument >0); (2) bring to one monotone function and read off the argument inequality (flip if base <1); (3) intersect with the domain; (4) for $|\log|$ or mixed signs, split cases before step 2. Parametric problems: solve for x with the parameter as a constant, then discuss the parameter ranges where the solution set changes (usually at the boundary points where a log argument hits $0/1$ or a base equals 1).

Number-of-digits and order-of-magnitude models

$d(N)=\lfloor \lg N\rfloor+1$ (see sheet 4). Variants: how many integers $n\leq N$ satisfy something? take logs; how many powers 2^k lie in $[a,b]$? count k with $\lg a\leq k\lg 2<\lg b$.

WORKED EXAMPLE

Worked. How many digits does 3^{50} have? First two digits estimate.
$\lg(3^{50})=50\lg 3\approx 23.856$. So $d=\lfloor 23.856\rfloor+1=24$ digits. For the leading digits: $10^{0.856}\approx 7.18$, so $3^{50}\approx 7.18\times 10^{23}$ — leading digit 7. (Check: $3^{50}=717897987691852588770249$ ✓ 24 digits, starts with 7 ✓.)

Growth models (the syllabus-fit applied framing)

Populations, compound interest, and radioactive decay all use $N(t)=N_0a^t$ or N_0e^{kt}: with a fixed percentage growth per period the multiplier is constant and the index is time. Doubling time for e^{kt}: $kt=\ln 2$, so $t=\ln 2/k$ (with $\ln 2\approx 0.693$). Half-life is the same with $-\ln 2$. These are **models built from the supplied function** a^x — the exam task is always to translate words into a^x or $\log$, then solve with the toolkit above.

Comparison of growth classes (use as ordering, not as calculus): as $x\to\infty$ (any fixed bases $a>1$, $b>1$, $c>1$): $\log_c x \ll x^{1/n}$ (for every n) $\ll a^x$. "Slowest to fastest: log, then root/power, then exponential." This is why compound growth overtakes polynomial growth eventually, and why taking logs linearises exponential comparisons.

SHORT DERIVATION

Solving $a^x = b^{x+c}$ — the canonical parameter problem

Take ln: $x\ln a = (x+c)\ln b \Rightarrow x(\ln a - \ln b) = c\ln b \Rightarrow x = \dfrac{c\ln b}{\ln a - \ln b} = \dfrac{c}{\log_b a - 1}$,

provided $a \neq b$ (if $a = b$: either no solution ($c \neq 0$) or all x ($c = 0$)). Every "two-exponential crossing" problem is this shape after factoring.

COMMON TRAP

Common traps — applications edition

- Word problems smuggle in domains: "$\log$ of the population" assumes the population is positive; "after t years" may mean $t \in \mathbb{N}$ vs $t \geq 0$ real.
- Doubling/half-life formulas use **natural** logs; mixing $\lg$ and $\ln$ in one computation is a classic arithmetic error.
- $\lfloor \lg N \rfloor + 1$ counts digits of the **integer part** of N; for non-integers state the convention used.
- Growth "per cent per year" compounds: 1.06^t, not $1 + 0.06t$ (that linear model is a different, weaker statement).

LESSON 07 / 08

The whole toolkit on one sheet

Exponents ($a>0$ convention). $a^r a^s = a^{r+s}$; $(a^r)^s = a^{rs}$ (needs $a>0$); $(ab)^r = a^r b^r$ (needs $a, b>0$); $a^{-r} = 1/a^r$; $a^0 = 1$ (empty product, incl. $0^0 = 1$ as convention). Rational p/q in lowest terms; negative base only with odd q. $\sqrt{x^2} = |x|$.

Exponential function. Domain $\mathbb{R}$, range $(0, \infty)$, through $(0, 1)$, asymptote $y = 0$; $a > 1$ increasing, $0 < a < 1$ decreasing; $f(x+y) = f(x)f(y)$; $a^x = e^{x \ln a}$.

Logarithm. $\log_a y = x \Leftrightarrow a^x = y$; domain $y > 0$, base $a > 0$, $a \neq 1$; laws $\log MN = \log M + \log N$, $\log(M/N) = \log M - \log N$, $\log M^p = p \log M$ (with $M > 0$; repair $\log x^2 = 2\log|x|$); change of base $\log_a b = \frac{\ln b}{\ln a}$; chain $\log_a b \cdot \log_b c = \log_a c$; inverses $a^{\log_a x} = x$ $(x > 0)$, $\log_a(a^x) = x$.

Signs and monotonicity. $\log_a x > 0 \Leftrightarrow (x-1)(a-1) > 0$; $\log_a$ increasing for $a > 1$, decreasing for $0 < a < 1$ (flip inequalities); $\lg 2 \approx 0.301$, $\lg 3 \approx 0.477$, $\ln 2 \approx 0.693$, $e \approx 2.718$.

Equations. Types: common base ($f = g$), quadratic in a^x ($t = a^x > 0$), collapse-logs-then-exponentiate (domain first!), $x^{\log x}$ (take logs). Always: domain, solve, verify in the original.

Applications. Digits $d = \lfloor \lg N \rfloor + 1$; compare magnitudes by $\ln$; a^b vs b^a via $\frac{\ln t}{t}$ (max at e); growth $N_0 a^t$ / $N_0 e^{kt}$, doubling $t = \ln 2/k$; growth hierarchy log $\ll$ power $\ll$ exponential.

KEY RESULT

Five questions to ask any exponent/log problem. (1) What is the domain (bases > 0, arguments > 0, bases $\neq 1$)? (2) Which single monotone function can I bring everything under? (3) Is the base > 1 or < 1 (inequality direction)? (4) After solving, which candidates survive the original equation? (5) Are units/bases mixed ($\lg$ vs $\ln$, degrees vs real exponents)?

ASSUMPTIONS & CONVENTIONS

Conventions fixed for the whole chapter: a^x real-valued for $a > 0$ (negative bases only through integer exponents or odd roots); $0^0 = 1$ as empty-product convention; $\lg = \log_{10}$, $\ln = \log_e$; all logs real, so arguments positive; "log" with no base means a formula holds for every legal base.

STANDARD APPLICATION

Where this toolkit reappears. Domain/monotonicity discipline powers calculus limits (1^∞ forms via ln), analysis of $f(t) = \frac{\ln t}{t}$, and counting estimates (digits, digits of $n!$, order of magnitude) in later chapters and in Olympiad number theory (Legendre's formula — see Part II). The habit "convert a^x to $e^{x \ln a}$" is the bridge to differential calculus.

COMMON TRAP

Final trap sweep

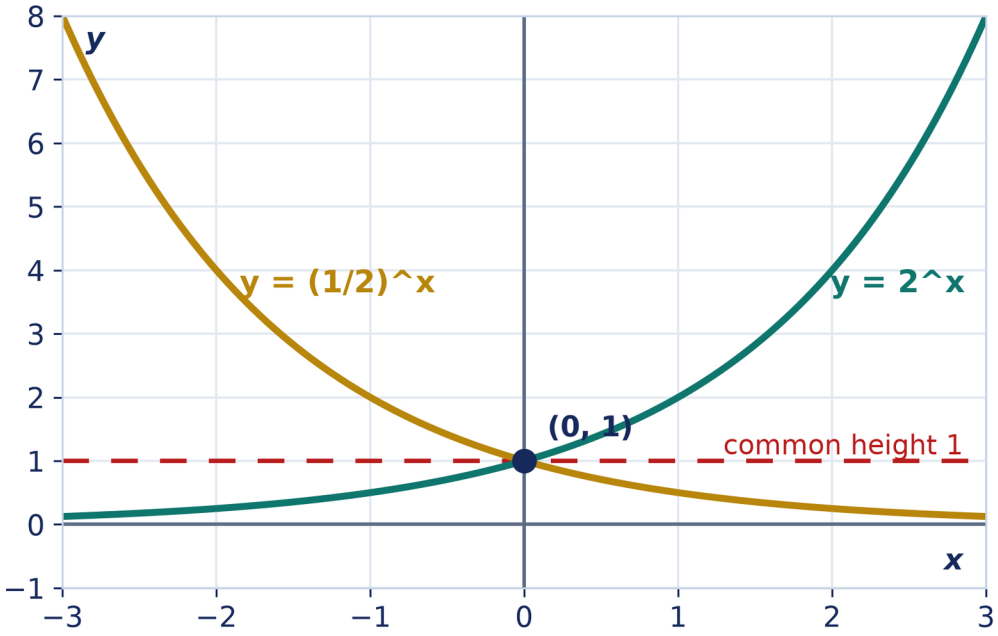
- $\log_a(M+N)$ has **no** expansion; $(\log M)^p \neq \log M^p$.
- $(a^r)^s = a^{rs}$ and $(ab)^r = a^r b^r$ silently need $a, b > 0$.
- Inequality flips when the base is in $(0, 1)$.
- Ghost roots from squaring / collapsing logs: verify in the original.
- $2^x = c$ has a real solution only for $c > 0$; then $x = \log_2 c$ uniquely.

LESSON 08 / 08

The base changes the direction of growth

For $a > 0$ and $a \neq 1$, $f(x) = a^x$ has domain $\mathbb{R}$, range $(0, \infty)$, and passes through $(0, 1)$. If $a > 1$, it increases; if $0 < a < 1$, it decreases. In both cases the x-axis is a horizontal asymptote, but the axis is never reached at a finite x.

Growing and decaying exponentials are mirror images



READ THE FIGURE
Since 0.5^x = 2^(-x), the decay graph is obtained by reflecting the growth graph in the y-axis. Neither curve crosses the x-axis.

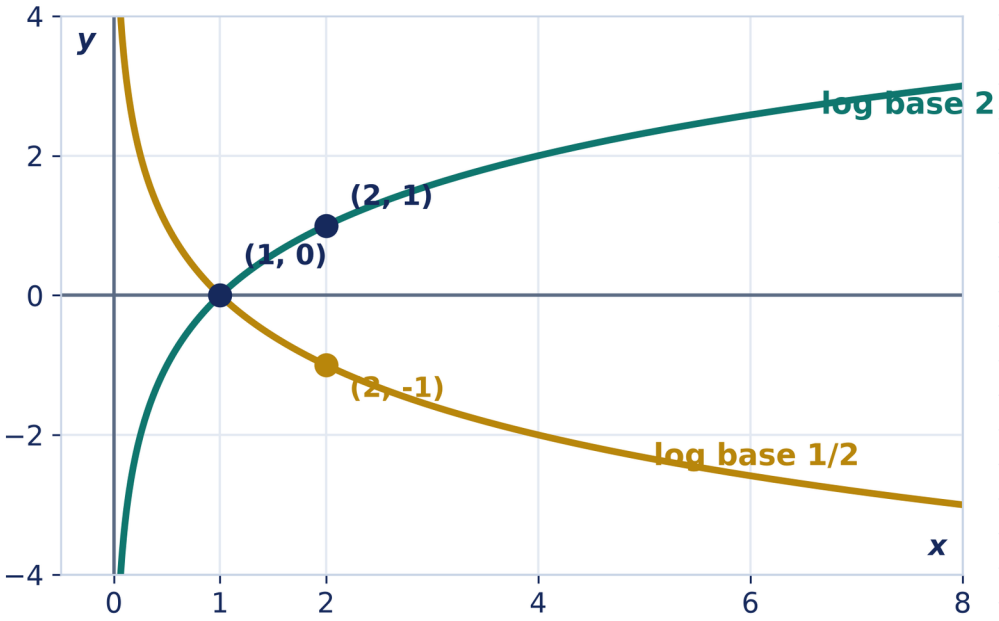
The reflection identity $a^{-x} = (1/a)^x$ explains the whole picture. It also gives a quick comparison tool: for $a > 1$, $a^u > a^v$ exactly when $u > v$; for $0 < a < 1$, the inequality reverses. When an exponential inequality has a variable base, first split into these base cases instead of applying a memorised rule blindly.

A logarithm undoes the exponential

The equation $y = \log_a x$ means $a^y = x$. Consequently the logarithm has domain $(0, \infty)$ and range $\mathbb{R}$. Its graph passes through $(1, 0)$ because $a^0 = 1$, and through $(a, 1)$ because $a^1 = a$. The vertical line $x = 0$ is an asymptote.

Logarithmic shape carries the laws

Logarithmic shape for bases greater and less than one



READ THE FIGURE
Every logarithm graph lies to the right of $x = 0$. Changing a base $a > 1$ to its reciprocal reflects the graph in the x-axis.

The laws are therefore not arbitrary algebraic rules. Since $a^u a^v = a^{u+v}$, multiplication of inputs becomes addition of their logarithms. The hypothesis $M, N > 0$ is essential: $\log_a(MN)$ can exist when both M, N are negative, while $\log_a M$ and $\log_a N$ are not real. Expand only after writing the domain.

A graph cannot replace the domain filter

Solve $\log_2(x-1) + \log_2(x-3) = 3$. Before combining logs, require $x - 1 > 0$ and $x - 3 > 0$, hence $x > 3$. On that domain,

$$\log_2((x-1)(x-3)) = 3 \quad \Longrightarrow \quad (x-1)(x-3) = 8.$$

Thus $x^2 - 4x - 5 = 0$, so $x = 5$ or $x = -1$. Only 5 survives the original domain.

Candidate roots are not yet solutions

The original domain selects the admissible root



READ THE FIGURE

The quadratic supplies candidates; the logarithm conditions decide which candidate is allowed. Substitution at x = 5 gives log2(4) + log2(2) = 3.

COMMON TRAP

Never repair an invalid input by simplifying harder

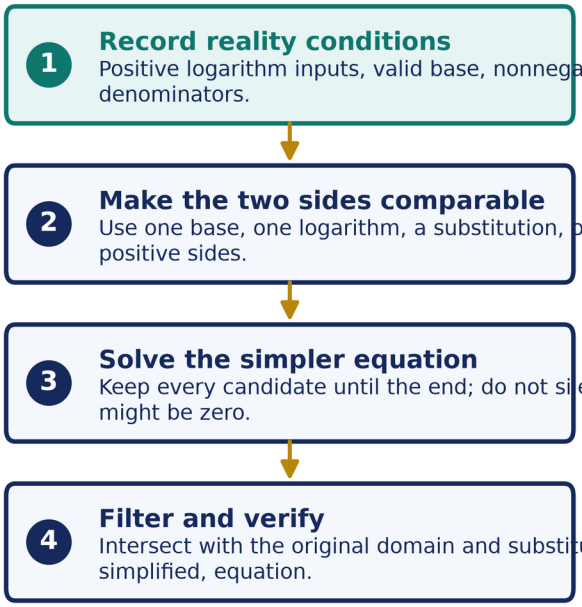
At $x=-1$, the combined product $(x-1)(x-3)=8$ is positive, so the combined logarithm exists. The original two logarithms nevertheless do not exist over $\mathbb{R}$. This is why the domain must be copied from the original equation before any law is used.

Choose the method from the structure

A substitution $t=a^x$ is useful only with $t>0$. A common-base step is useful only when the base is valid and both sides are defined. Taking a logarithm of both sides requires positive sides. Each method has an admission ticket.

Exponential and logarithmic equations: a reliable order

Domain first, algebra second, verification last



READ THE FIGURE

Most wrong answers in this chapter come from skipping the first or last step, not from the algebra in the middle.

WORKED EXAMPLE

A substitution with two branches

For $4^x - 5\cdot 2^x + 4 = 0$, put $t = 2^x$, so $t > 0$ and $4^x = t^2$. Then $t^2 - 5t + 4 = 0$, giving $t = 1$ or $t = 4$. Both are positive, so $2^x = 1$ gives $x = 0$ and $2^x = 4$ gives $x = 2$. Direct substitution gives $1 - 5 + 4 = 0$ and $16 - 20 + 4 = 0$. The condition $t > 0$ is part of the substitution, not an afterthought.

KEY RESULT

What the pictures should leave behind

An exponential never returns a nonpositive output; a logarithm never accepts a nonpositive input. Growth reverses when the base is replaced by its reciprocal. Graphs reveal sign, monotonicity and asymptotes, while the original domain remains the final authority for solutions.

Worked synthesis: growth models become logarithm models

An exponential model carries both a starting value and a time unit. For $N(t) = 80 \cdot 3^{t/5}$, the initial value is $N(0) = 80$ and the multiplication by 3 occurs every 5 time units. To find when $N(t) = 720$, divide before taking a logarithm:

$$3^{t/5} = 9 = 3^2 \quad \Longrightarrow \quad t = 10.$$

The equivalent logarithmic calculation is $t = 5\log_3 9 = 10$. Substitution gives $80 \cdot 3^2 = 720$, verifying both the value and the time unit.

WORKED EXAMPLE

Decay uses the same equation, not a new trick

For $Q(t) = 64(1/2)^{t/3}$, solve $Q(t) = 4$. Then $(1/2)^{t/3} = 1/16 = (1/2)^4$, so $t = 12$. A negative exponential rate would produce the same conclusion after logarithms; the essential hypotheses are a positive base different from 1 and a positive target-to-initial ratio.

COMMON TRAP

Do not take a logarithm before isolating the exponential

In an equation such as $ab^{kt} + c = d$, first obtain $b^{kt} = (d - c)/a$ and check that the right side is positive. Only then is the real logarithm step available.

KEY RESULT

Interpret the answer

Report what the computed t measures, check whether the model asks for elapsed time or calendar time, and substitute into the original model. A numerically correct exponent with the wrong unit is not a correct model solution.

LESSON 01 / 02

Status of this sheet

The equation $x^y = y^x$ and multi-base logarithmic systems are **standard IOQM/RMO algebra** and appear in specialised exam questions; everything below is derived from the Part I toolkit (monotonicity, change of base, $\frac{\ln t}{t}$) with no new theory assumed. Nothing here "can never appear in JEE" — the classification is only that these models go beyond the current syllabus sentence "Logarithms and their properties".

$x^y = y^x$ — the complete positive solution set

For $x, y > 0$: if $x = y$ everything works. If $x \neq y$, take ln: $y\ln x = x\ln y$, i.e. $\dfrac{\ln x}{x} = \dfrac{\ln y}{y}$.

So x, y are two distinct points where $f(t) = \frac{\ln t}{t}$ takes the **same value** — and f increases on $(1, e]$, decreases on $[e, \infty)$, so any level is cut at most twice.

SHORT DERIVATION

Parametrisation of all unequal solutions

Put $y = tx$ with $t > 0$, $t \neq 1$. Then $x^{tx} = (tx)^x \Rightarrow x^t = tx \Rightarrow x^{t-1} = t \Rightarrow x = t^{1/(t-1)}$, $y = t^{t/(t-1)}$. Writing $t = 1 + \dfrac{1}{n}$ gives the classic family $x = \left(1 + \dfrac{1}{n}\right)^n$, $y = \left(1 + \dfrac{1}{n}\right)^{n+1}$ for $n > 0$ ($n \in \mathbb{Q}^+$ or $\mathbb{N}$). Check $n = 1$: $(x, y) = (2, 4)$: $2^4 = 4^2 = 16$ ✓. Integer solutions with $x \neq y$: exactly $(2, 4)$ and $(4, 2)$.

WORKED EXAMPLE

Worked. Real solutions with $x < y$, $xy = 8$, $x^y = y^x$?

$\frac{\ln x}{x} = \frac{\ln y}{y}$ with $y = 8/x$: solve $\frac{\ln x}{x} = \frac{\ln(8/x)}{8/x}$ numerically/graphically by monotonicity arguments; or use the parametrisation: $\frac{y}{x} = t = 1 + \frac{1}{n}$, $xy = 8$ fixes n. From $y = tx$ and $xy = 8$: $tx^2 = 8$. With $x = t^{1/(t-1)}$ this becomes $t^{1 + \frac{2}{t-1}} = 8 \Rightarrow t^{\frac{t+1}{t-1}} = 8$. Trial $t = 4$: $4^{5/3} \approx 10.08 \neq 8$; $t = 2$: $2^3 = 8$ ✓. So $t = 2$, $x = 2$, $y = 4$. Answer: $(x, y) = (2, 4)$ (and the symmetric pair if order is free).

Multi-base logarithmic systems

The three moves: (1) reciprocity $\log_a b \cdot \log_b a = 1$; (2) chain $\log_a b \cdot \log_b c = \log_a c$; (3) change every base to ln (or to a common base c) and clear denominators.

WORKED EXAMPLE

Worked. Solve $\log_2 x = \log_x 4$

Domain $x > 0$, $x \neq 1$. *Reciprocity:* $\log_x 4 = \dfrac{\ln 4}{\ln x} = \dfrac{2\ln 2}{\ln x}$; *also* $\log_2 x = \dfrac{\ln x}{\ln 2}$. *Set* $u = \ln x$: $\dfrac{u}{\ln 2} = \dfrac{2\ln 2}{u} \Rightarrow u^2 = 2(\ln 2)^2 \Rightarrow u = \pm \ln 2 \cdot \sqrt{2}$. *So* $x = 2^{\sqrt{2}}$ *or* $x = 2^{-\sqrt{2}}$. *Check both are* > 0 *and* $\neq 1$ ✓. *Answer:* $x \in \{2^{\sqrt{2}},\ 2^{-\sqrt{2}}\}$.

Towers obey a convention: $a^{b^c} = a^{(b^c)}$ — *exponentiation is* ***right-associative***; $(a^b)^c = a^{bc} \neq a^{b^c}$ *in general. Nested logs peel from the inside:* $\log_2(\log_3(\log_4 256)) = \log_2(\log_3 4) = \log_2\left(\dfrac{\ln 4}{\ln 3}\right)$ — *leave as exact expressions; never invent a "log of log" law.*

COMMON TRAP

Traps on this sheet

- $x^y = y^x$ *over* ***integers*** *vs* ***reals*** *give different answer sets; the parametrisation covers reals; integers need* $t = 1 + 1/n$ *with* t *rational* $\Rightarrow n = 1$ *only among integers (for* $x < y$ *positive integers:* $\{2, 4\}$*).*
- $\log_a b = \log_a c \Rightarrow b = c$ ***needs the same base***; $\log_2 x = \log_3 x \Rightarrow x = 1$ *only.*
- *Solving* $\log_x(\cdot)$ *adds base conditions* $x > 0$, $x \neq 1$ *on top of argument positivity.*
- $f(t) = \frac{\ln t}{t}$ *is* ***not*** *monotone on* $(0, \infty)$ — *it turns at* $t = e$*; two solutions for* $f(t) = c \in (0, 1/e)$ *sit on opposite sides of* e.

Assumptions & conventions

All variables positive reals unless stated; integer solutions are a separate reading of the same equations; $\frac{\ln t}{t}$ *analysis uses its stated monotonicity shape (derivable later by calculus; here taken as the known shape of the standard curve, with the level-curve picture as the geometric model).*

LESSON 02 / 02

Status of this sheet

Order-of-magnitude estimates and Legendre's formula are ***classic IOQM/RMO number theory*** *built from the Part I toolkit (logs, floors, geometric progressions). The hierarchy statements below are comparisons between standard functions, models derivable from supplied information — not additional theory claimed as a syllabus item.*

The growth hierarchy (state it as ordering, use it as a tool)

For fixed $a>1$, $b>1$, $c>1$, $k\in\mathbb{N}$, *as* $x\to\infty$:

$$\log_c x \ll x \ll x^k \ll a^x \ll b^{b^x}.$$

Use in estimates: *to compare* $n^{\log n}$*-shaped quantities, take* $\log$ *once or twice until everything is polynomial. Example: which is larger for huge* n *—* 100^n *or* n^{100}*? Take* $\ln$*:* $n\ln 100$ *vs* $100\ln n$*; divide by* n*:* $\ln 100\approx 4.605$ *vs* $\frac{100\ln n}{n}\to 0$*. So* 100^n *wins for large* n*; at* $n=100$ *the two are* ***equal****, for* $n>100$ *the exponential* 100^n *is larger, for* $e<n<100$ *the power* n^{100} *is larger (crossover where* $\frac{\ln n}{n}=\frac{\ln 100}{100}=0.046$*).*

KEY RESULT

Digit-counting engine. $d(N)=\lfloor \lg N\rfloor+1$ *for integers* $N\geq 1$*. Sums of digits in base* b*:* $N=\sum d_i b^i$ *with* $0\leq d_i<b$*; the digit-sum* $s_b(N)=\sum d_i$ *satisfies* $N-s_b(N)$ *is divisible by* $b-1$ *(casting out* $b-1$*'s). This pair (*$\lfloor\log\rfloor$*, digit sum) is exactly what Legendre's formula packages.*

Legendre's formula — the exponent of a prime in $n!$

For prime p*:* $v_p(n!)=\displaystyle\sum_{k=1}^{\infty}\left\lfloor\frac{n}{p^k}\right\rfloor=\frac{n-s_p(n)}{p-1}$*, where* $s_p(n)$ *is the digit sum of* n *in base* p *(finitely many non-zero terms: stop when* $p^k>n$*).*

SHORT DERIVATION

Derivation — count multiples, then clean up with digits

Among $1,2,\ldots,n$*, exactly* $\lfloor n/p\rfloor$ *are divisible by* p*; among those,* $\lfloor n/p^2\rfloor$ *contribute a* ***second*** *factor* p*; and so on — each integer* m *contributes* $v_p(m)$ *factors, so the total is* $\displaystyle\sum_{k\geq 1}\lfloor n/p^k\rfloor$ ✓ *(count each* m *once per power of* p *dividing it). For the closed form, write* $n=s_p(n)+(p-1)\displaystyle\sum_{k\geq 1}\lfloor n/p^k\rfloor$ *(base-*p *identity:* $\lfloor n/p^k\rfloor=\displaystyle\sum_{j>k} d_j p^{j-k-1}$*), rearranged to the formula.*

WORKED EXAMPLE

Worked. $v_2(100!)$ and $v_5(100!)$

$v_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97$. *Via digits:* $100 = (1100100)_2$, $s_2 = 3$, $\frac{100-3}{1} = 97$ ✓. $v_5(100!) = 20 + 4 = 24$; *digits:* $100 = (400)_5$, $s_5 = 4$, $\frac{100-4}{4} = 24$ ✓. *Consequence:* $100!$ *ends in exactly* 24 *zeros (the power of* $10 = 2 \cdot 5$ *is the smaller of* v_2, v_5*).*

Series estimates with logs and floors

Truncation errors: $\sum_{k=1}^{n} \frac{1}{k}$ *grows like* $\ln n$ *(harmonic estimate:* $\ln(n+1) \le H_n \le 1 + \ln n$ *— provable by comparing the sum with* $\int \frac{dx}{x}$*, a standard derived model). The telescoping estimate* $\frac{1}{k+1} \le \ln\left(1 + \frac{1}{k}\right) \le \frac{1}{k}$ *(from* $\frac{1}{1+t} \le \ln(1+t)$*-style log bounds at* $t = 1/k$*) is the workhorse behind it.*

WORKED EXAMPLE

Worked. Bound $H_{100} = 1 + \frac{1}{2} + \cdots + \frac{1}{100}$

$\ln 101 \approx 4.615 \le H_{100} \le 1 + \ln 100 \approx 5.605$; *the true value is* ≈ 5.187 ✓ *inside the bracket. Use such brackets to decide sign/size questions without computing sums.*

COMMON TRAP

Traps on this sheet

- $\lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \cdots$ *is* **not** *a geometric series — do not sum it as one; each term is a floor.*
- *Trailing zeros of* $n!$ *are* $v_5(n!)$ *(5s are rarer than 2s), not* $v_2(n!)$*.*
- log *estimates compare growth rates; a bound valid "for large* n*" must not be quoted for a specific small* n *without checking (e.g.* $n^{100} > 100^n$ *fails for* $n = 2$*).*
- *Digit sums* s_p *depend on the base;* $s_2(100) = 3$ *counts* ***ones in binary****, not decimal digits.*

Assumptions & conventions

p *prime;* $v_p(m)$ *= exact exponent of* p *in* m*; floors are real floors; harmonic estimates quoted as standard derived bounds (checked numerically above); hierarchy statements are eventual comparisons (there is always a crossover point).*